NUTS & BOLTS — COMPLETE LIST

Apollo Track Pod · two pods, one per side · nothing below class 8.8 anywhere · grade 8 (US) = class 10.9

REV 011
2026-08-22
source: FASTENERS.md + apollo_track_pod_rev011.scad

A · PIVOT AXLE

Sheet 2 · §7.2 / §7.3

ITEM	/POD	BOTH
M16 × 195 hex bolt, part-threaded 45 mm one end, cl.10.9 (the pivot axle both arms swing on; smooth shank 150 must span all 4 bushings + spacer — bushings never ride on thread; head stamped 10.9)	1	**2**
M16 nylock nut (closes the pivot stack — one per bolt only, it is a headed bolt; zero end-float then back ⅛ turn, no torque figure)	1	**2**
Ø16 hardened flat washer (under the pivot bolt head and under the nylock)	2	**4**
Ø16 × Ø30 × 1.5 hardened thrust washer (between every steel face that rotates against another in the pivot stack: spacer–flange, fork–fork scissor gap, flange–sleeve)	6	**12**

Substitute if the bolt can't be found: M16 double-end stud × 190 + 2 nylocks per pod, smooth middle ≥ 150. Prefer 8.8/B7 over stainless. Companions in this stack (not fasteners): 4 flanged bronze bushings SAE 841 Ø16×Ø22×20 per pod.

B · IDLER WHEEL AXLES

supersedes §7.5 · studs purchased 2026-07-25

ITEM	/POD	BOTH
GB901 M12 × 110 double-end stud, 8.8 (TRAILING idler axle — runs in the 25 mm tensioner slot)	1	**2**
GB901 M12 × 120 double-end stud, 8.8 (LEADING idler axle — plain Ø15.4 bore; longer because the leading fork is wider)	1	**2**
M12 nylock nut (two per stud — a double-end stud has no head)	4	**8**
M12 washer, ≈2.5 thick, snug Ø13 bore (caps the Ø15 sleeve column AND clamps the plate — goes directly against the plate, ALWAYS before the eye nut; a sloppy bore that slips over the Ø15 ring lets the fork crush onto the wheel)	4	**8**

Sleeves (inner tube 50.8 TR / 66.5 LD + 2 × 6.35 rings per axle) come from the 500 mm Ø15×Ø12 pipe in the §7.1 steel order — cut long, file square, fit at dry-stack. Pass test: wheel spins free with zero side-play at full torque; if it drags, the tube is short — never back the nut off instead.

C · FORK MOUNT

§7.7

ITEM	/POD	BOTH
M8 × 30 bolt, cl.10.9 (bolts each carrier to its fork leg: carrier 6 + leg 4 + nut; locks the torque-arm rotation)	2	**4**
M8 nylock nut (on the fork-leg bolts)	2	**4**
M10 hub-motor axle nut (holds the hub motor in the carrier's flatted Ø10.4 torque-arm hole — REUSED from the donor pod, do not buy)	—	reuse

D · SHOCKS

§7.7 / §9.5 · 4 shocks total, eyes measured Ø8

ITEM	/POD	BOTH
M8 part-threaded bolt, ≈135 (lower shock eye — through both arm plates at station a; the eye rides the spacer sleeve on the smooth shank)	2	**4**
M8 nylock nut (on the lower shock bolts)	2	**4**
M8 pin/bolt ≈35–40 + nylock (UPPER shock eye, through the Ø8.4 hole in the carrier tab — missing from §7.7; length unspecified anywhere, measure at the §9.5 fitting with the real shock eye in hand)	2	**4**

Companion (steel order, not a fastener): Ø15 OD × Ø9 ID × ≈45 spacer sleeve for each lower eye — 4 total. Confirm the ≈135 lower length at §9.5 too.

E · BELT TENSIONER

§7.6 · Rev 004a hardware · trailing arm only

ITEM	/POD	BOTH
M6 × 60 full-thread draw bolt, cl.8.8+ (threads into the eye nut through the welded lug; advancing it draws the axle rearward in the slot — loaded in tension; advance both sides evenly)	2	**4**
M6 jam nut (locks the draw bolt against the lug's rear face once belt tension is set)	2	**4**
M6 lifting eye nut, DIN 582-style, eye inner Ø20 (rides the protruding trailing-axle end, clamped washer–eye–nylock; the draw bolt pulls it — nothing on the pod gets drilled or tapped. Owner already has these)	2	**4**

F · THREADED CONSUMABLES

§7.8

ITEM	/POD	BOTH
M6 grease zerk (one per fork boss if you add the recommended cross-drilled grease passage — verify the thread really is M6)	1	**2**
Nylock + hardened washer assortment, M8/M10, class 10 / Gr8 (for the fitting-up you will inevitably redo)		1 box
Blue (medium) threadlocker + paint pen (threadlocker on non-nylock threads; pen for torque witness marks on every fastener)		1 ea

DO NOT BUY — SUPERSEDED BY REV 011

~~M8 threaded rod ≈172 + 4 nuts + washers (§7.6 keel standoff)~~	keel is deleted — fork legs box the carriers
~~M8 keel bolt through the carriers~~	Rev 011 carrier strips are 220 with no keel hole
~~4 × M15×1.5 fine nuts + ~8 Ø15 washers (§7.5)~~	idler axles are now M12 studs; Ø15 shaft kept only as spare
~~Second M16 nylock per pivot bolt~~	the bolt has a head — one nut per bolt
~~Anything below class 8.8~~	unmarked hardware-store bolts are out, everywhere

TORQUE

M6 (draw bolts / jam nuts)	snug + jam nut — sets belt tension, not preload	
M8 cl.10.9	`30 N·m`	
M10 (hub axle, reused nuts)	`60 N·m`	
M12 (idler axle nylocks)	`105 N·m`	— only after the wheel spins free with zero side-play
M16 (pivot nylock)	zero end-float, then back off ⅛ turn — running fit, never a torque figure	

Paint-pen witness marks on everything. Retorque at 1 h / 5 h / then every 20 h (§10).

SHOPPING CHECKLIST — BOTH PODS

☐ M16 × 195 part-thread cl.10.9 bolt	2
☐ M16 nylock nut	2
☐ Ø16 hardened flat washer	4
☐ Ø16×Ø30×1.5 thrust washer	12
☐ M12 × 110 stud 8.8 (have — purchased)	2
☐ M12 × 120 stud 8.8 (have — purchased)	2
☐ M12 nylock nut	8
☐ M12 washer, snug Ø13 bore	8
☐ M8 × 30 cl.10.9 bolt	4

☐ M8 part-thread bolt ≈135	4
☐ M8 pin/bolt ≈35–40 (measure at §9.5 first)	4
☐ M8 nylock nut	12
☐ M6 × 60 full-thread draw bolt 8.8	4
☐ M6 jam nut	4
☐ M6 eye nut DIN 582 Ø20 (have — owner's)	4
☐ M6 grease zerk	2
☐ M10 hub axle nut (reuse from donor)	–

Assembly order that matters: on the idler axles the M12 washer goes directly against the arm plate, then the eye nut (trailing only), then the nylock — washer first is what caps the Ø15 sleeve column. Reading §B's stack as "eye against the plate" leaves the column uncapped and the fork can be crushed onto the wheel.